

Program : Diploma in Robotic Process Automation	
Course Code : 3301	Course Title: Problem Solving Using Python
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L: 3 T: 0 P: 0)	Periods per semester: 45

Course Objectives:

- To learn and understand the basics of Python programming.
- Develop different programming applications using Python.
- To learn the concept of file handling and exception handling.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic functions and features of Computer, Operating system and Internet applications.		Introduction to IT Systems	1
Basic programming skills in C language.		Introduction to Computation and Programming	2

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop algorithmic solution to simple problems and construct Python programs using control statements and expressions.	11	Applying

CO2	Develop programs using functions and slicing operation.	14	Applying
CO3	Use Python data structures – List, Tuple, and Dictionary – for solving problems.	9	Applying
CO4	Explain files, exceptions, and modules in Python for solving problems	9	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	3	3				
CO2	3	3	3				
CO3	3	3	3				
CO4	3	3	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop algorithmic solution to simple problems and construct Python programs using control statements and expressions.		
M1.01	Explain the methodology of problem solving - algorithm, flow chart, and pseudo code.	1	Understanding.
M1.02	Illustrate the basic concept of a Python- identifiers, keywords, variables and data types.	2	Understanding
M1.03	Explain operators and expressions- Expressions, Arithmetic, comparison, assignment, logical, bitwise, membership and identity operators, precedence of operators, associativity and input output functions.	3	Understanding
M1.04	Explain Control statements- if elif else statement, while statement, for statement, break and continue statements.	3	Understanding

M1.05	Develop programs using Control statements.	2	Applying
Contents: Problem Analysis: - Methodology of problem solving- Algorithm and flowchart with examples, pseudo code. Introduction to Python: - Identifiers, keywords, variables, data types. Operators, expressions, precedence of operators, associativity. Input and output functions. Control statements: if elif else statement, iteration statements- while, for. Break and continue statements. Programming examples.			
CO2:	Develop programs using functions and slicing operation.		
M2.01	Illustrate the concept function, types of functions – built-in-functions, type conversion, type coercion, mathematical functions	2	Understanding.
M2.02	Explain the concept of user defined functions- function definition, function calling, function returning values	3	Understanding
M2.03	Illustrate the difference between actual and formal parameters. Types of formal arguments- required, default and keyword arguments.	2	Understanding.
M2.04	Explain the concept of scope of variable-local and global scope and recursive function.	1	Understanding.
M2.05	Develop python programs using functions	2	Applying.
M2.06	Summarize the concept of string operations- concatenation, repetition, membership and slicing operations.	2	Understanding
M2.07	Develop programs using the concept of string operations.	2	Applying
	Series Test – I	1	
Contents: Functions: - Built-in-functions: -Type conversion, type coercion, Mathematical functions. User defined functions: - Function definition, function calling, function returning values. Arguments and parameters, required arguments, default arguments, keyword arguments. Recursive functions. Scope of variables-local and global scope. Programming examples. Strings: - Indexing, string operations- concatenation, repetition, membership and slicing. Built-in-functions. Programming examples.			

CO3:	Use Python data structures – List, Tuple, and Dictionary – for solving problems.		
M3.01	Explain the concept of Lists- accessing elements in List and operations- concatenation, repetition, and in operator	1	Understanding
M3.02	Illustrate the concept of traversing a list and built-in-List methods-len(), append(), insert() etc.	1	Understanding
M3.03	Develop programs using List	1	Applying
M3.04	Explain the concept of Tuples- creation, accessing values in Tuples and operations- concatenation, repetition, and in operator	1	Understanding
M3.05	Summarize built-in-Tuple functions- len(), max(), min() etc. – with examples.	1	Understanding
M3.06	Develop programs using Tuples	1	Applying
M3.07	Explain how dictionaries are created, accessing values in dictionary and traversing operations	1	Understanding
M3.08	Summarize built-in-dictionary functions with examples	1	Understanding
M3.09	Develop programs using Dictionary	1	Applying

Contents:

Lists: -

Introduction: - values and accessing elements, List operations- concatenation, repetition and in operator. Traversing a List. Built-in List methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. Programming examples.

Tuples: -

Introduction: - creating Tuples, accessing values in Tuples, operations- concatenation, repetition and in operator. Built-in-Tuple functions- len(), max(), min() etc. Programming examples.

Dictionaries:-

Introduction: - creating Dictionary, accessing values in Dictionary. Traversing. Built-in-functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. Programming examples.

CO4:	Explain files, exceptions, and modules in Python for solving problems		
M4.01	Explain the concept of files and types of files. Perform the operations on text, binary and csv files such as opening, reading, writing, closing and setting offsets in a file.	2	Understanding
M4.02	Explain the concept of pickle module with examples	1	Understanding
M4.03	Develop programs for performing the operations on	2	Applying

	files		
M4.04	Explain the concept of exceptions and built-in-exceptions	1	Understanding
M4.05	Explain how exceptions are handled with examples.	1	Understanding
M4.06	Develop programs for handling exceptions.	2	Applying
	Series Test – II	1	

Contents:

Files: -

Introduction: - Types of files- Text, binary and CSV files. Opening, closing, reading and writing in a text binary and CSV files. Setting offsets in a file. Pickle module- load() and dump(). Programming examples

Exceptions: -

Exceptions, built-in-exceptions, handling exceptions – Programming examples.

Text / Reference

T/R	Book Title/Author
T1	Balagurusamy E, Introduction to Computing and Problem Solving Using Python.
R1	Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015.
R2	Perkovic .L., Introduction to Computing Using Python, 2nd Edition
R3	Zelle. J., Python Programming: An Introduction to Computer Science.
R4	Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist : learning with Python, 1st Edition

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/python/index.htm
2	https://www.javatpoint.com/python-tutorial
3	https://www.programiz.com/python-programming
4	https://www.w3schools.com/python/default.asp